Subquery

Przemysław Starosta - 2026

Subquery – one row

	first_name	last_name	department_id
1	Matthew	Weiss	50
2	Adam	Fripp	50
3	Payam	Kaufling	50
4	Shanta	Vollman	50
5	Kevin	Mourgos	50
6	Julia	Nayer	50

```
select
     first_name,
     last_name,
     department_id
from employees
where
          department_id =
          (select department_id from departments  where department_name = 'Shipping')
```

Subquery – one row - error

Results | Messages

```
Msg 512, Level 16, State 1, Line 1
Subquery returned more than 1 value. This is not permitted when the subquery
follows =, !=, <, <= , >, >= or when the subquery is used as an expression.
```

```sql
select
	first_name,
	last_name,
	department_id
from employees
where
	department_id =
	(select department_id from departments
	where department_name in ('Shipping','Administration'))
```

Subquery – more than one row

40	Samuel	McLeod	50
41	Vance	Jones	50
42	Alana	Walsh	50
43	Kevin	Feeney	50
44	Donald	OConnell	50
45	Douglas	Grant	50
46	Jennifer	Whalen	10

```
select
     first_name,
     last_name,
     department_id
from employees
where
     department_id IN (select department_id from departments
          where department_name in
          ('Shipping','Administration'))
```

Subquery – many in one query

	first_name	last_name	department_id
1	David	Williams	60
2	Valli	Jackson	60
3	Diana	Nguyen	60
4	Nandita	Sarchand	50
5	Alexis	Bull	50
6	Sarah	Bell	50
7	Jennifer	Whalen	10

```
select
    first_name,
    last_name,
    department_id
    from employees
where salary between (select min_salary from jobs
                                    where job_title = 'Programmer')
          and (select max_salary from jobs
                                    where job_title = 'Stock Clerk')
```

Subquery – in having

	department_id	(No column name)
1	NULL	7000.00
2	10	4400.00
3	40	6500.00
4	70	10000.00

```
select
    department_id,
    sum(salary)
    from employees
group by department_id
having sum(salary) < (select sum(salary) from employees
                            where department_id = 20)
```

Subquery – operator ANY

	first_name	last_name	salary	department_id
1	Steven	King	24000.00	90
2	Neena	Yang	17000.00	90
3	Lex	Garcia	17000.00	90
4	Alexander	James	9000.00	60
5	Nancy	Gruenberg	12008.00	100
6	Daniel	Faviet	9000.00	100
7	John	Chen	8200.00	100
8	Ismael	Sciarra	7700.00	100

```sql
select
    first_name,
    last_name,
    salary,
    department_id
    from employees
where salary >ANY (select salary from employees
                            where department_id = 20)
```

Subquery – operator ALL

	first_name	last_name	salary	department_id
1	Steven	King	24000.00	90
2	Neena	Yang	17000.00	90
3	Lex	Garcia	17000.00	90
4	John	Singh	14000.00	80
5	Karen	Partners	13500.00	80

```
select
    first_name,
    last_name,
    salary,
    department_id
    from employees
where salary >ALL (select salary from employees
                            where department_id = 20)
```

Subquery – correlation

	first_name	last_name	salary	department_id
1	Michael	Martinez	13000.00	20
2	Den	Li	11000.00	30
3	Nandita	Sarchand	4200.00	50
4	Alexis	Bull	4100.00	50
5	Kelly	Chung	3800.00	50
6	Jennifer	Dilly	3600.00	50
7	Sarah	Bell	4000.00	50

```
select
    first_name,
    last_name,
    salary,
    department_id
    from employees e
where salary >  (select avg(salary) from employees
                where department_id = e.department_id)
```

Subquery – correlation - EXISTS

	first_name	last_name	salary	department_id
1	Matthew	Weiss	8000.00	50
2	Adam	Fripp	8200.00	50
3	Payam	Kaufling	7900.00	50
4	Shanta	Vollman	6500.00	50
5	Kevin	Mourgos	5800.00	50
6	Julia	Nayer	3200.00	50
7	Irene	Mikkilineni	2700.00	50
8	James	Landry	2400.00	50

```
select
    first_name,
    last_name,
    salary,
    department_id
    from employees e
where EXISTS  (select * from departments
where department_name = 'Shipping’  and department_id = e.department_id)
```

Subquery – correlation – NOT EXISTS

	first_name	last_name	salary	department_id
1	Steven	King	24000.00	90
2	Neena	Yang	17000.00	90
3	Lex	Garcia	17000.00	90
4	Alexander	James	9000.00	60
5	Bruce	Miller	6000.00	60
6	David	Williams	4800.00	60

```
select
    first_name,
    last_name,
    salary,
    department_id
    from employees e
where EXISTS  (select * from departments
where department_name = 'Shipping’  and department_id = e.department_id)
```

Problem

```
Messages
Msg 207, Level 16, State 1, Line 6
Invalid column name 'full_salary'.
```

```sql
select
    first_name,
    last_name,
    salary + (salary * isnull(commission_pct,0)) full_salary
    from employees e
where full_salary > 15000
```

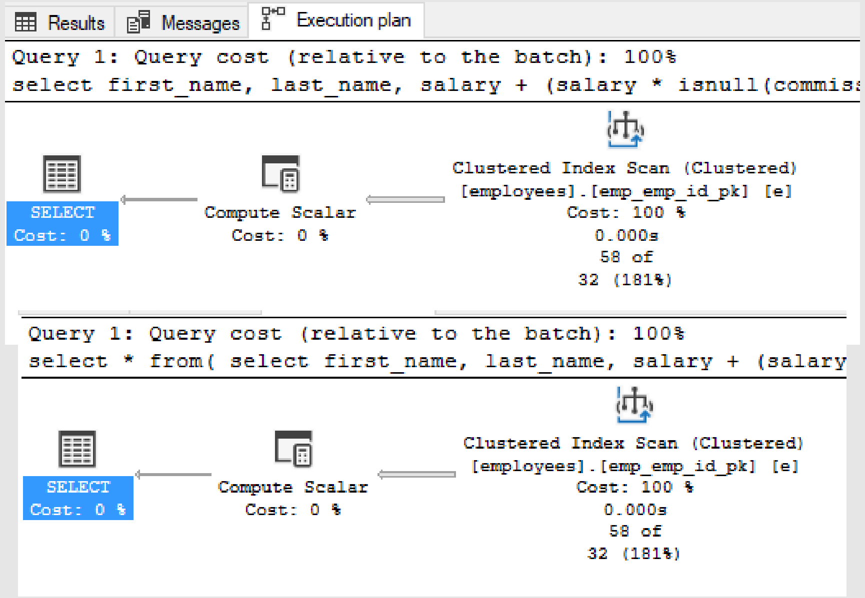
Problem – resolve by subquery

	first_name	last_name	full_salary
1	Steven	King	24000.0000
2	Neena	Yang	17000.0000
3	Lex	Garcia	17000.0000
4	John	Singh	19600.0000
5	Karen	Partners	17550.0000
6	Alberto	Errazuriz	15600.0000

```
select * from(
    select
        first_name,
        last_name,
        salary + (salary * isnull(commission_pct,0)) full_salary
        from employees e ) t1
where t1.full_salary > 15000
```

Compare execution plans

```
select
        first_name,
        last_name,
        salary + (salary * isnull(commission_pct,0))
full_salary
        from employees e
where salary + (salary * isnull(commission_pct,0)) > 15000
```




```
select * from(
    select
        first_name,
        last_name,
        salary + (salary * isnull(commission_pct,0)) full_salary
        from employees e ) t1
where t1.full_salary > 15000
```

Subquery in select

	first_name	last_name	dep_name
1	Steven	King	Executive
2	Neena	Yang	Executive
3	Lex	Garcia	Executive
4	Alexander	James	IT
5	Bruce	Miller	IT
6	David	Williams	IT
7	Valli	Jackson	IT
8	Diana	Nguyen	IT
9	Nancy	Gruenberg	Finance

```sql
select
        first_name,
        last_name,
        (select department_name from departments where department_id =
                e.department_id) dep_name
        from employees e
```

Subquery in from

	first_name	last_name
1	Jennifer	Whalen

```
select
        t1.first_name,
        t1.last_name
        from (select first_name, last_name, department_id
                from employees
                where department_id =10) t1
```

WITH

Określa tymczasowy nazwany zestaw wyników, znany jako tabela przechodnia(CTE – Common Table Expresion). Pochodzi z prostego zapytania i jest zdefiniowany w zakresie wykonywania pojedynczej instrukcji SELECT, INSERT, UPDATE, DELETE lub MERGE.

WITH

	department_name	last_name
1	Executive	King
2	Executive	Yang
3	Executive	Garcia
4	IT	James
5	IT	Miller
6	IT	Williams

```
WITH
        q1(department_name, last_name)
AS
 (SELECT d.department_name, e.last_name FROM employees e JOIN
        departments d on e.department_id=d.department_id)
SELECT * FROM q1
```

Many WITH

	department_name	last_name	job_title	last_name
1	Executive	King	President	King
2	Sales	King	President	King
3	Executive	Yang	Administration Vice President	Yang
4	Executive	Garcia	Administration Vice President	Garcia
5	IT	James	Programmer	James
6	IT	Miller	Programmer	Miller

```
WITH q1(department_name, last_name) AS
 (SELECT d.department_name, e.last_name FROM employees e JOIN departments d
on   e.department_id=d.department_id),
q2(job_title, last_name) AS
 (SELECT j.job_title, e.last_name FROM employees e JOIN jobs j on
    e.job_id=j.job_id)
SELECT * FROM q1 q join q2 q2 on q.last_name=q2.last_name
```

Rekursywne WITH

	ManagerID	EmployeeID	EmployeeLevel
1	NULL	100	0
2	100	101	1
3	100	102	1
4	100	114	1
5	100	120	1
6	100	121	1
7	100	122	1
8	100	123	1

```sql
WITH Report(ManagerID, EmployeeID, EmployeeLevel) AS
(
    SELECT manager_id, employee_id, 0 AS EmployeeLevel
    FROM employees e1
    WHERE manager_id IS NULL
    UNION ALL
    SELECT e.manager_id, e.employee_id, EmployeeLevel + 1
    FROM employees AS e JOIN Report AS r ON e.manager_id = r.EmployeeID
)
SELECT ManagerID, EmployeeID, EmployeeLevel
FROM Report
ORDER BY ManagerID
```